

AMMAN ARAB UNIVERSITY

جامعة عمان العربية

College of Information Technology

كلية تكنولوجيا المعلومات



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Enterprise Resource Planning System

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Amman - Jordan
2025/2026

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1.1 Problem Statement and project Scope

Many businesses that handle buying and hiring use manual processes or systems that don't work together, which causes mistakes and inefficiencies. Spreadsheets or paper-based methods are often used to manage suppliers, keep track of products, and process purchase orders in buying. This can lead to data that doesn't match up, delays, and not being able to see inventory in real time.

In the same way, manually keeping track of employee records, attendance, and salary information can lead to duplicate data, mistakes, and trouble getting the information you need when you need it.

Not having a single system that connects purchasing and HR functions makes things even harder, like not being able to keep track of which employees are in charge of which purchasing tasks. This lack of coordination makes it harder to make decisions and run things efficiently.

So, there needs to be a centralized ERP system that makes buying and HR processes easier, makes data more accurate, and boosts productivity in the workplace.

Scope:

- 1)Add, update, or delete information about suppliers.
- 2)Control data about products and categories
- 3)Make and keep track of purchase orders
- 4)Keep track of what you bought and how much of it you bought
- 5)Automatically change the amount of stock in your inventory
- 6)Keep track of which employee made each purchase order.

1.2 Project Plan and Schedule

The ERP system will be built in phases, with a structured approach to make sure that planning, building, and testing go smoothly.

Here is the Gantt Chart to schedule and manage the ERP system. And how we distribute the tasks of this project.

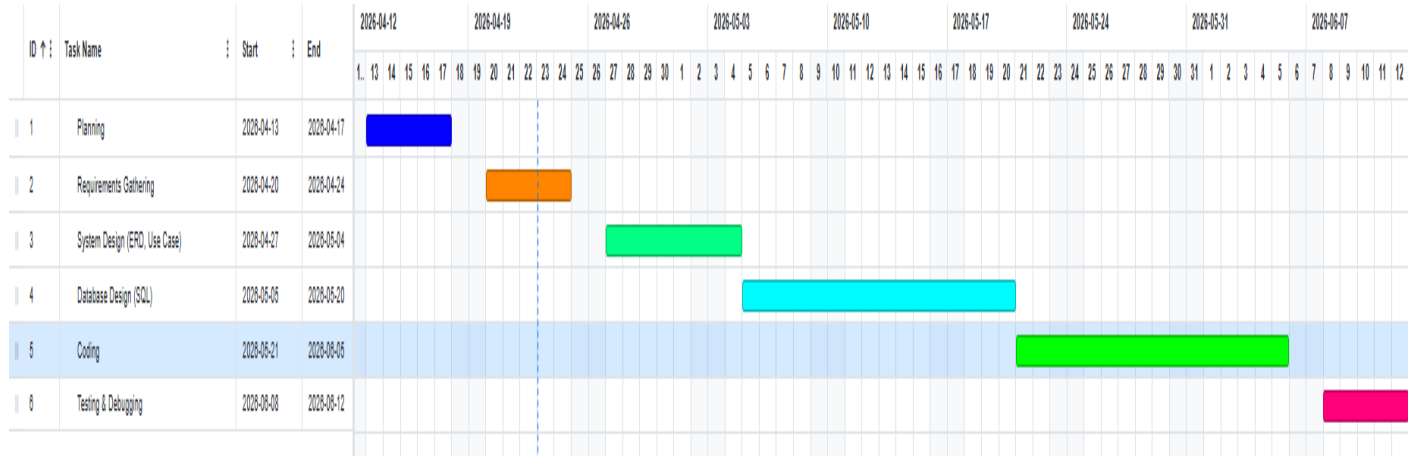


Figure 1: Gantt Chart of project

1.3 2. System Analysis

These are the Requirements we chose for this project.

2.1 Functional Requirements

- **FR1. Supplier Management**
The system shall allow users to add, update, view, and delete supplier information, including name, contact details, and address.
- **FR2. Product Management**
The system shall enable users to manage product records, including product name, category, price, and available stock quantity.
- **FR3. Purchase Order Creation**
The system shall allow users to create purchase orders by selecting a supplier and adding one or more products with their respective quantities and prices.
- **FR4. Employee Association**
The system shall record the employee responsible for creating each purchase order to ensure accountability and traceability.
- **FR5. Inventory Update**
The system shall automatically update product stock levels when a purchase order is confirmed or completed.

2.2 Use Case Diagram

The following Use Case Diagram illustrates the main interactions between the system and its users, showing how different actors engage with the system's functionalities.

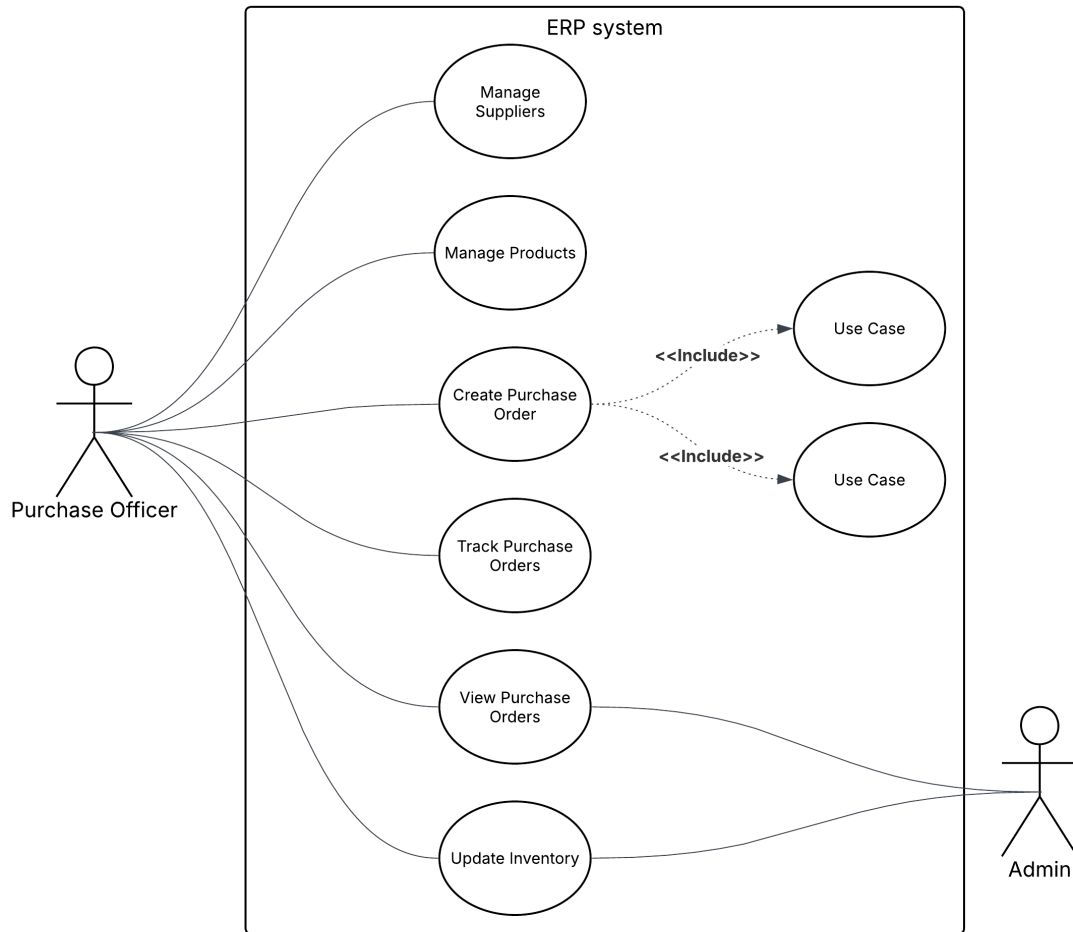


Figure 2: Use Case diagram of system

3. System Design

The following Entity-Relationship Diagram (ERD) represents the structure of the database, showing the main entities, their attributes, and the relationships between them.

3.1 ER-Diagram

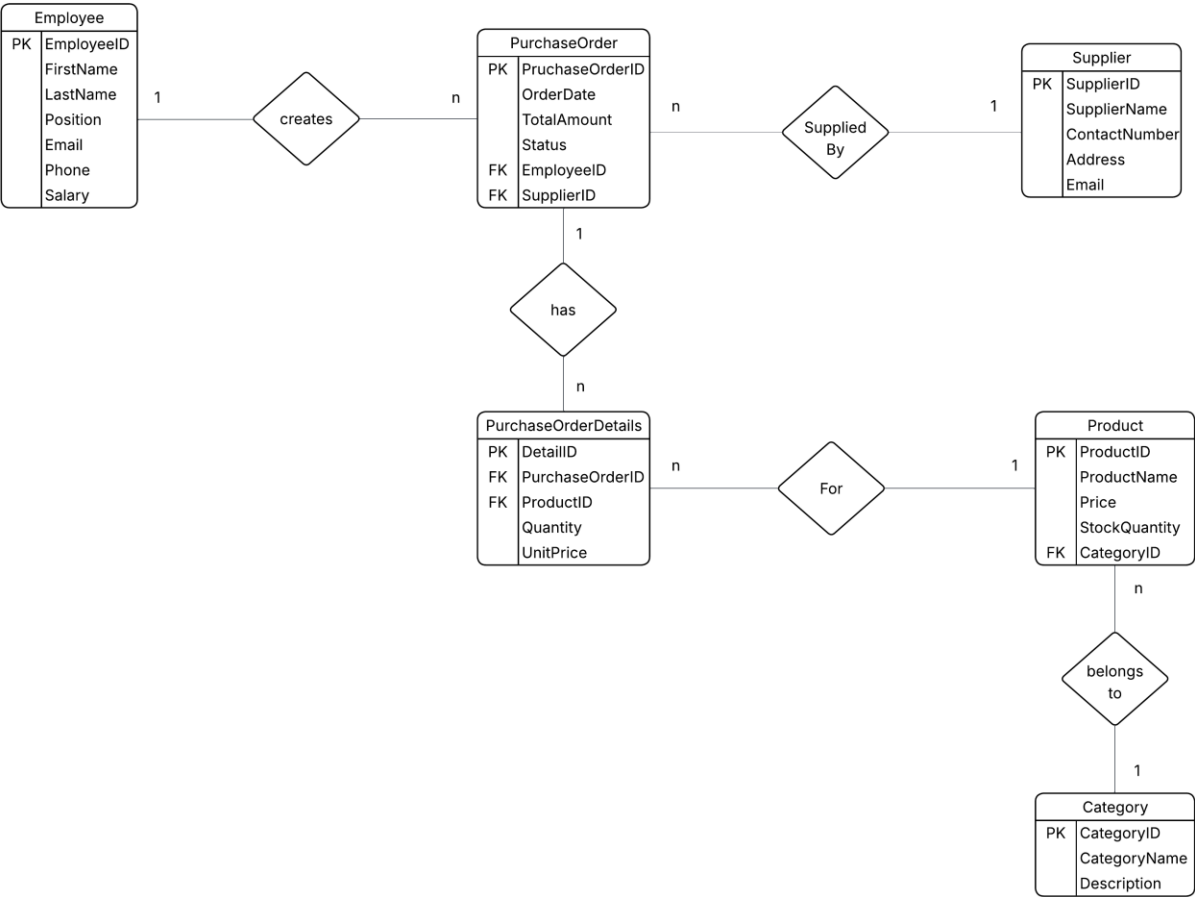
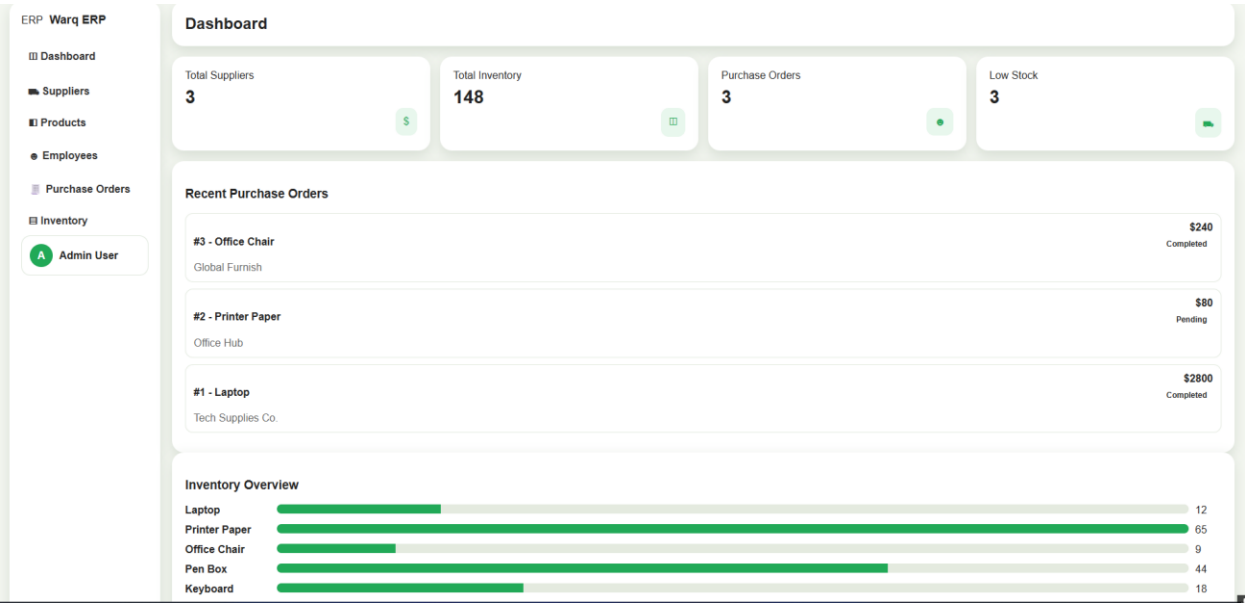


Figure 3: ER diagram of System

3.2 User Interface (prototype)



ERP Warq ERP

Dashboard

Suppliers

Products

Employees

Purchase Orders

Inventory

A

Admin User

Products

Product Management

Product Name

Category

Electronics

Price

Stock Quantity

0

0

Save Product

Clear

ID	Product	Category	Price	Stock	Actions
5	Keyboard	Electronics	\$25	18	Edit Delete
4	Pen Box	Stationery	\$4	44	Edit Delete
3	Office Chair	Furniture	\$120	9	Edit Delete
2	Printer Paper	Office	\$8	65	Edit Delete
1	Laptop	Electronics	\$700	12	Edit Delete

Employees

Employee Management

First Name

Last Name

Position

Email

Phone

Salary

0

Save Employee

Clear

ID	Name	Position	Email	Phone	Salary	Actions
3	Lina Khaled	Inventory Manager	lina@erp.com	0793333333	\$950	Edit Delete
2	Abdullah Madaat	HR Officer	abdullah@erp.com	0792222222	\$850	Edit Delete
1	Anwar Maslamani	Purchase Officer	anwar@erp.com	0791111111	\$900	Edit Delete

ERP Warq ERP

Dashboard

Suppliers

Products

Employees

Purchase Orders

Inventory

A

Admin User

Purchase Orders

Create Purchase Order

Supplier

Global Furnish

Responsible Employee

Lina Khaled

Product

Keyboard - Stock: 18

Quantity

1

Status

Completed

Create Order

ID	Date	Supplier	Employee	Product	Qty	Total	Status	Actions
#3	2026-03-05	Global Furnish	Abdullah Madaat	Office Chair	2	\$240	Completed	Delete
#2	2026-02-08	Office Hub	Lina Khaled	Printer Paper	10	\$80	Pending	Delete
#1	2026-01-15	Tech Supplies Co.	Anwar Maslamani	Laptop	4	\$2800	Completed	Delete

Inventory

Inventory Tracking

ID	Product	Category	Current Stock	Price	Status
5	Keyboard	Electronics	18	\$25	Low Stock
4	Pen Box	Stationery	44	\$4	Available
3	Office Chair	Furniture	9	\$120	Critical
2	Printer Paper	Office	65	\$8	Available
1	Laptop	Electronics	12	\$700	Low Stock